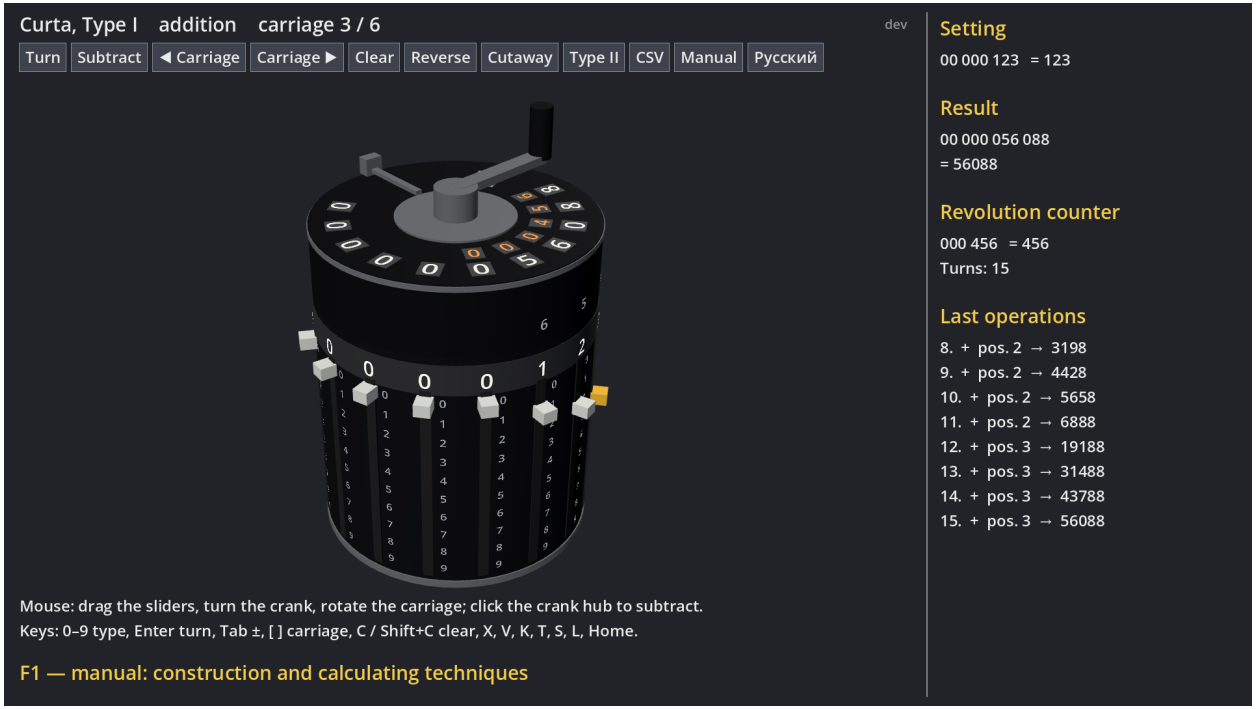
# The Curta mechanical calculator

Interactive 3D model: construction, controls and calculating techniques




The model after the 123 × 456 example (section 6). F1 opens/closes this manual, ← / → flip pages.

# 1. History

The Curta is the smallest four-function mechanical calculator ever made in series. You hold the pepper-grinder-sized cylinder in your left hand and turn the crank on top with your right. Hence its nicknames: the “pepper grinder” and the “math grenade”.

It was invented by the Viennese engineer **Curt Herzstark** (1902–1988), son of a calculating-machine factory owner. He had been thinking about a pocket machine since the late 1930s. In 1943 he was arrested and sent to the Buchenwald concentration camp; there he was put to work in the camp workshop and allowed to draw the calculator. By the liberation in 1945 the drawings were almost complete.

After the war production was set up by Contina AG in Liechtenstein. **Type I** appeared in 1948, the larger **Type II** in 1954. They were made until the early 1970s, when electronic calculators took over; about 140 thousand were built according to various sources. Rally navigators, surveyors and engineers loved the Curta: fast, exact and with no batteries.

| | Type I | Type II |
|---|---|---|
| Setting sliders | 8 | 11 |
| Revolution counter | 6 | 8 |
| Result register | 11 | 15 |
| Carriage positions | 6 | 8 |
| Diameter × height, mm | 53 × 85 | 65 × 90 |

Switch the type with T in the program (all examples below are for Type I).

# 2. Parts of the machine

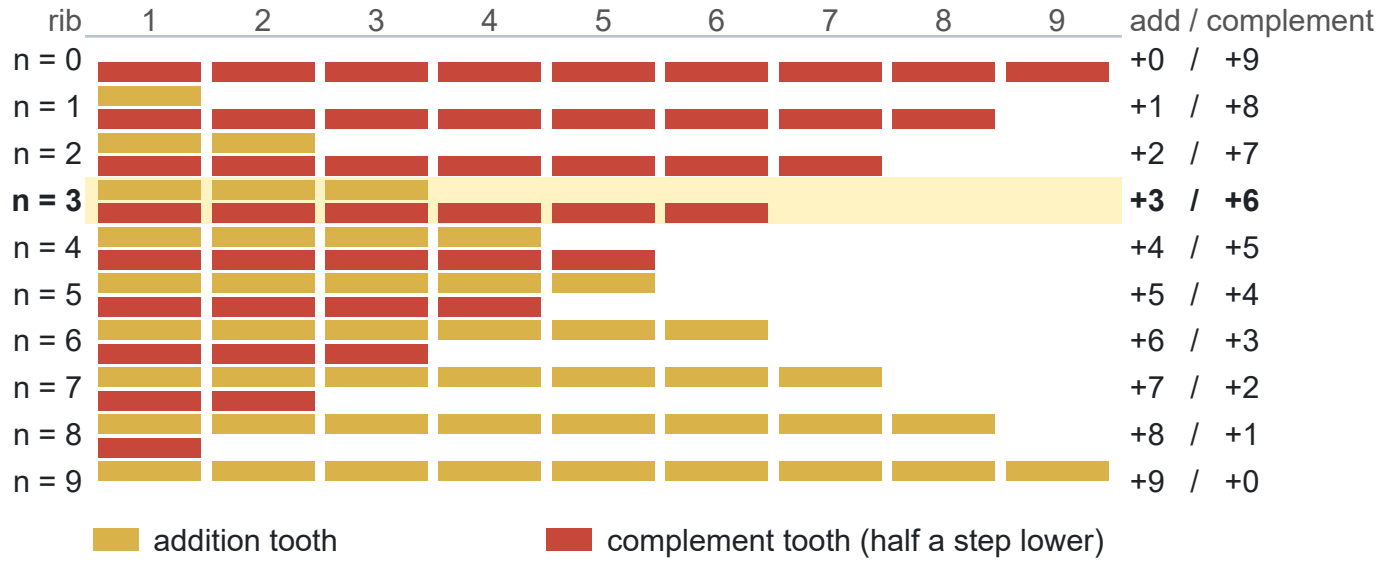
**Interlock.** While the crank is away from home you cannot shift the carriage, pull or push the crank, or use the clearing lever — the machine protects itself, and the model behaves the same way.

| Part | What it does |
|---|---|
| Setting sliders | Vertical slots on the body, one per digit; the highest digit on the left. 0 at the top, 9 at the bottom. The number on the sliders is the “setting”; it shows in the check window above the slots. |

| Part | What it does |
|---|---|
| Carriage | The top part; it rotates relative to the body one digit at a time. It carries two registers: the result (outer ring, white digits) and the revolution counter (inner ring, orange). The position number is on the carriage side next to the mark on the body. |
| Crank | One full clockwise turn adds the setting to the result and 1 to the counter. With the crank pulled up (red ring visible) a turn subtracts. The crank cannot turn backwards (ratchet). |
| Clearing lever | A ring lever on the carriage: sweep it a full circle to clear the result and the counter. |
| Counter reverse switch | Flips the counting sign: in division and square root subtractions count +1, so the counter shows the answer. |
| Stepped drum | The cylinder in the centre of the body, turning with the crank. Its teeth turn the slider pinions — see section 3. |

# 3. How the stepped drum works

The drum goes back to Leibniz in the 17th century: nine teeth of different lengths on a cylinder. The slider pinion slides along the drum; at level n exactly n teeth catch it per revolution, so the pinion and its result wheel turn by n tenths. The Curta has one drum for all digits: the pinions of all sliders stand around it.




The unrolled drum. A pinion at value 3 (yellow band) catches 3 teeth when adding, 6 when subtracting.

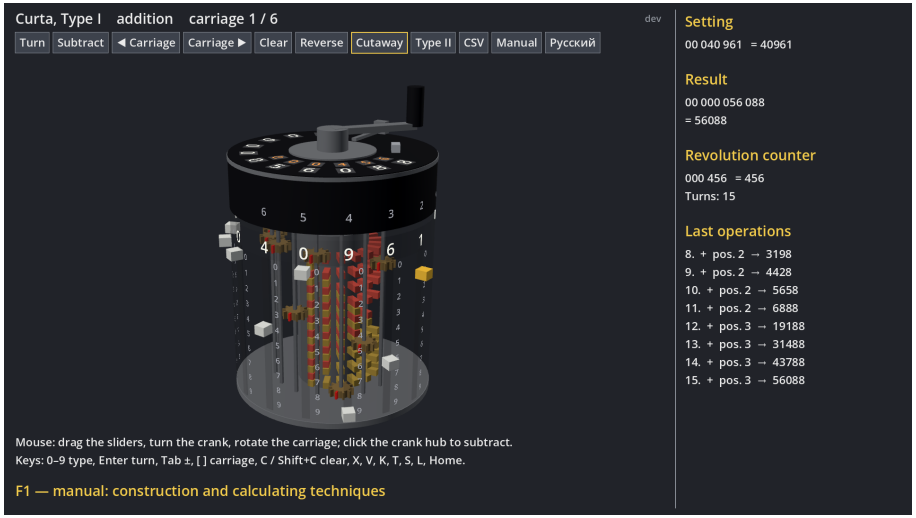
## Subtraction by adding the complement

Herzstark's key idea: between the addition teeth the drum carries **nines-complement** teeth — 9 − n at level n. Pulling the crank up lifts the drum half a step, and these teeth come under the pinions. One more "+1" tooth acts on the lowest digit under the carriage. Altogether $10^{w} - x$ is added, where w is the number of register digits and the carry out of the top digit is lost. So $R + (10^{w} - x) = R - x$: the machine subtracts without turning anything backwards.

Example: 0 − 1. Sliders 00000001, crank up. The drum gives each zero digit 9 teeth and the units 8 + 1 = 9 — the result is **99 999 999 999**, the counter 999 999. Nines at the top mean a negative number in complement form: −1. The program shows it with the signed value on the right panel.

## Tens carry

When a result wheel passes 9 → 0, a finger adds 1 to the next digit; the carry can ripple through a chain of nines. The carry out of the top digit is lost — which is exactly why complement subtraction works.




Cutaway (K): the body is transparent; the drum, the pinions on square shafts and their red marks are visible.

# 4. Program controls

| Mouse (and touch on Steam Deck) | Effect |
|---|---|
| Drag a slider knob up / down | Value 0–9; wheel over a slider — ±1 |
| Turn the crank clockwise | A full circle is one operation; it won’t go back; released just past home it snaps home |
| Click the crank hub | Pull up (subtract) / push in (add) |
| Rotate the carriage by its top | Shift by one digit |
| Sweep the clearing lever around | Clear result and counter |
| Click the switch on the carriage | Counter reverse (red = on) |
| Drag empty space / right button | Orbit the view; wheel — zoom |

## Keyboard

| Key | Action | Key | Action |
|---|---|---|---|
| 0–9, Backspace | Type a number (digits shift in from the right) | C | Clear registers (Shift+C — counter only) |
| ← / → | Select a slider | X | Sliders to 0 |
| ↑ / ↓ | Slider up / down (−1 / +1) | V | Counter reverse |
| Enter, Space | Turn the crank | K | Cutaway |
| Tab | Crank up / down | T | Type I / II |
| [ / ] | Carriage one position down / up | S | Log to CSV |
| Home | Reset view | L | Language |
| F1 | This manual (O opens the PDF) | | |

On the right is the register panel with signed values and the last operations. S saves the log user://curta_log.csv: n, op, carriage, setting, result, counter — one row per turn and per clear.

# 5. Addition and subtraction

## Example: 1234 + 5678 − 999.

**1.** Clear the registers (lever or C), carriage at position 1, crank pushed in.

**2.** Set 1234 and make a turn: the number goes to the result, counter 1.

**3.** Set 5678 and turn again — the sum is in the result.

**4.** Set 999, pull the crank up (hub or Tab) and turn — 999 is subtracted, the counter goes down.

| Step | Sliders | Crank | Result | Counter |
|---|---|---|---|---|
| 1 | 1,234 | add | 00 000 001 234 | 000 001 |
| 2 | 5,678 | add | 00 000 006 912 | 000 002 |
| 3 | 999 | subtract | 00 000 005 913 | 000 001 |

Answer: 5,913. The counter holds the “sum of turns” 1 + 1 − 1 = 1: not needed for addition, but in multiplication and division it gives the multiplier and the quotient.

# 6. Multiplication: turns and shifts

Multiplication is repeated addition, and shifting the carriage one position multiplies the setting by 10. So the multiplier is “cranked in” digit by digit: units at position 1, tens at position 2 and so on. The counter builds up the multiplier itself — a handy check that nothing went wrong.

## Example: 123 × 456.

| Carriage position | Multiplier digit | Turns | Result | Counter |
|---|---|---|---|---|
| 1 | 6 | 6 × + | 00 000 000 738 | 000 006 |
| 2 | 5 | 5 × + | 00 000 006 888 | 000 056 |
| 3 | 4 | 4 × + | 00 000 056 088 | 000 456 |

Answer: 56,088, the counter shows 456. 15 turns in total.

## Short-cut multiplication

Digits 7, 8, 9 are cheaper by subtraction: 99 = 100 − 1. For 123 × 99 two turns are enough instead of 18: one subtracting at position 1 and one adding at position 3.

| Position | Crank | Result | Counter |
|---|---|---|---|
| 1 | 1 × subtract | 99 999 999 877 | 999 999 |
| 3 | 1 × add | 00 000 012 177 | 000 099 |

The counter again shows 99: −1 at position 1 and +1 at position 3 make 100 − 1.

# 7. Division: subtract until overdraw

**1.** Clear the registers. Set the dividend 56090 and turn once at position 1 — it is now in the result. Clear only the counter: Shift+C.

**2.** Switch on the counter reverse (V): every subtraction now counts +1.

**3.** Set the divisor 123. Shift the carriage so the divisor's top digit is under the dividend's top digit: for 56090 and 123 — position 3.

**4.** Pull the crank up and turn until the result goes "negative" (nines appear at the top). That is one turn too many — undo it with one turn with the crank pushed in.

**5.** Shift the carriage one position down and repeat. After position 1 the counter holds the quotient, the result holds the remainder.

## Example: 56090 : 123.

| Position | Subtractions until overdraw | Undo | Quotient digit | Result | Counter |
|---|---|---|---|---|---|
| 3 | 5 | 1 × add | 4 | 00 000 006 890 | 000 400 |
| 2 | 6 | 1 × add | 5 | 00 000 000 740 | 000 450 |
| 1 | 7 | 1 × add | 6 | 00 000 000 002 | 000 456 |

Answer: 56090 = 123 × **456** + **2**. “Subtractions until overdraw” is one more than the quotient digit — the last turn and its undo cancel out in both the result and the counter.

For decimal places, shift the dividend left on the sliders (multiply by $10^k$) and remember where the decimal point goes.

# 8. Square root by odd numbers

Toepler's method (1865) relies on $1 + 3 + 5 + \ldots + (2n - 1) = n^2$. Subtract consecutive odd numbers while you can, and their count is the root. For a multi-digit root take two digits of the radicand per root digit: once part r of the root is known, subtract the odd numbers $20r + 1$, $20r + 3$, $20r + 5$, …

On the Curta each root digit has its own carriage position k, and the odd number is set on the sliders shifted by the same k digits. Together this gives the weight $100^k$ — exactly a pair of digits. With the counter reversed it collects the root digits. The extra subtraction is undone by one addition, as in division.

## Example: √15129 (pairs 1 | 51 | 29 → three root digits, positions 3, 2, 1).

| Position | Numbers subtracted (sliders) | Root digit | Result | Counter |
|---|---|---|---|---|
| 3 | 100, 300 | 1 | 00 000 005 129 | 000 100 |
| 2 | 210, 230, 250 | 2 | 00 000 000 729 | 000 120 |
| 1 | 241, 243, 245, 247 | 3 | 00 000 000 000 | 000 123 |

In red — the number that overdrew and was added back. Answer: √15129 = **123**, remainder 0.

Real Curta users don't reset the whole number each time: they move only the lowest slider by 2 (1 → 3 → 5 …), and when moving to the next digit they double it on the neighbouring slider — that is the 20r.